Understanding Late-Stage Breast Cancer Diagnosis in California

Exploring how socioeconomic conditions shape early detection outcomes for women under 50

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Extern — TruBridge Healthcare Data Analytics Project
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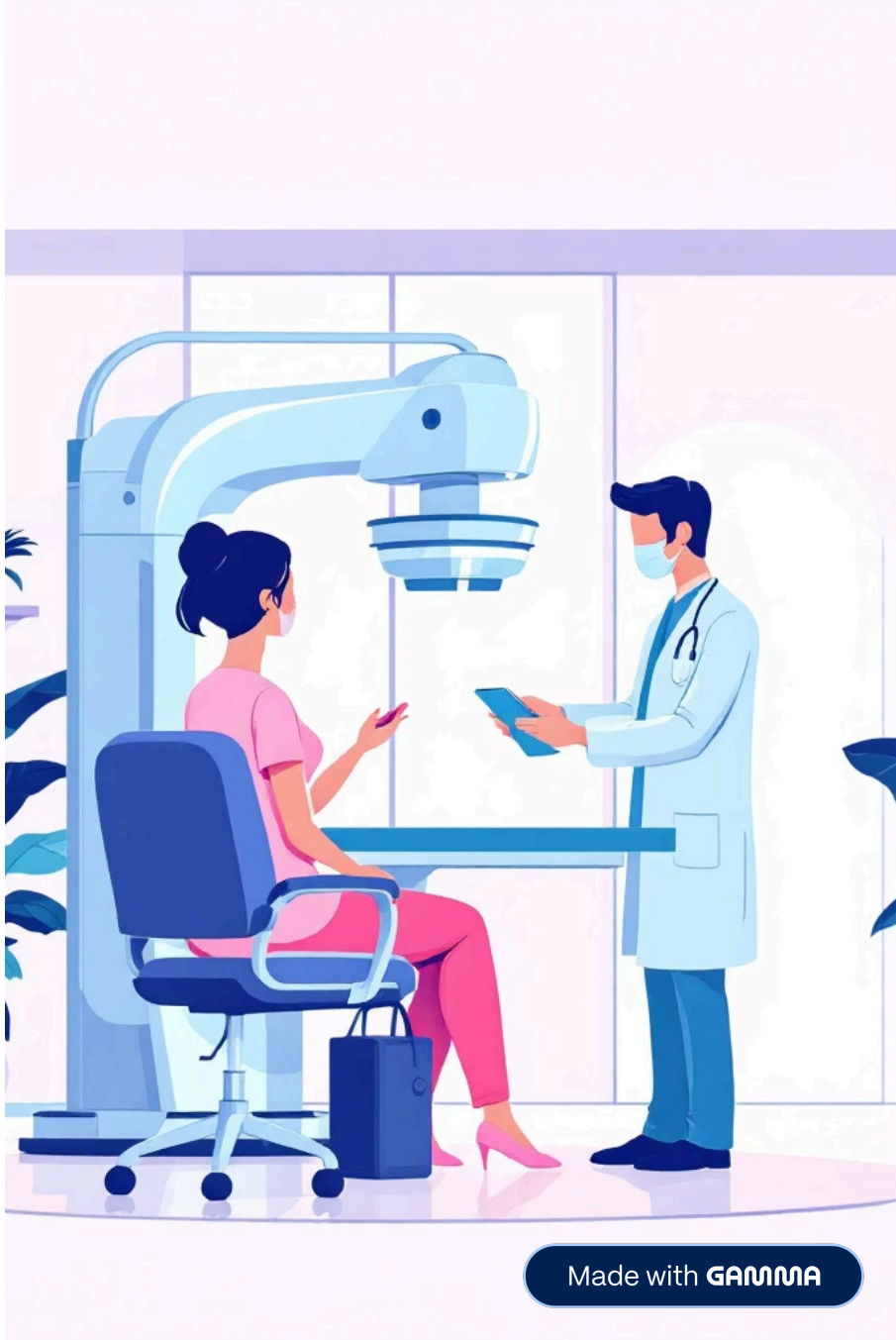

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The Problem & Research Question

> Women under 50 are often diagnosed at later stages of breast cancer, when treatment becomes more aggressive and survival outcomes worsen significantly. This critical gap in early detection points to underlying socioeconomic disparities.

Research Question: How are county-level socioeconomic factors—poverty rate, median income, and insurance coverage—associated with late-stage breast cancer diagnosis rates across California counties?

Data Sources: State Cancer Profiles merged with U.S. Census ACS data using county FIPS codes

18-21

Cases per 100,000

Most counties fall within this range for late-stage diagnosis rates among women under 50

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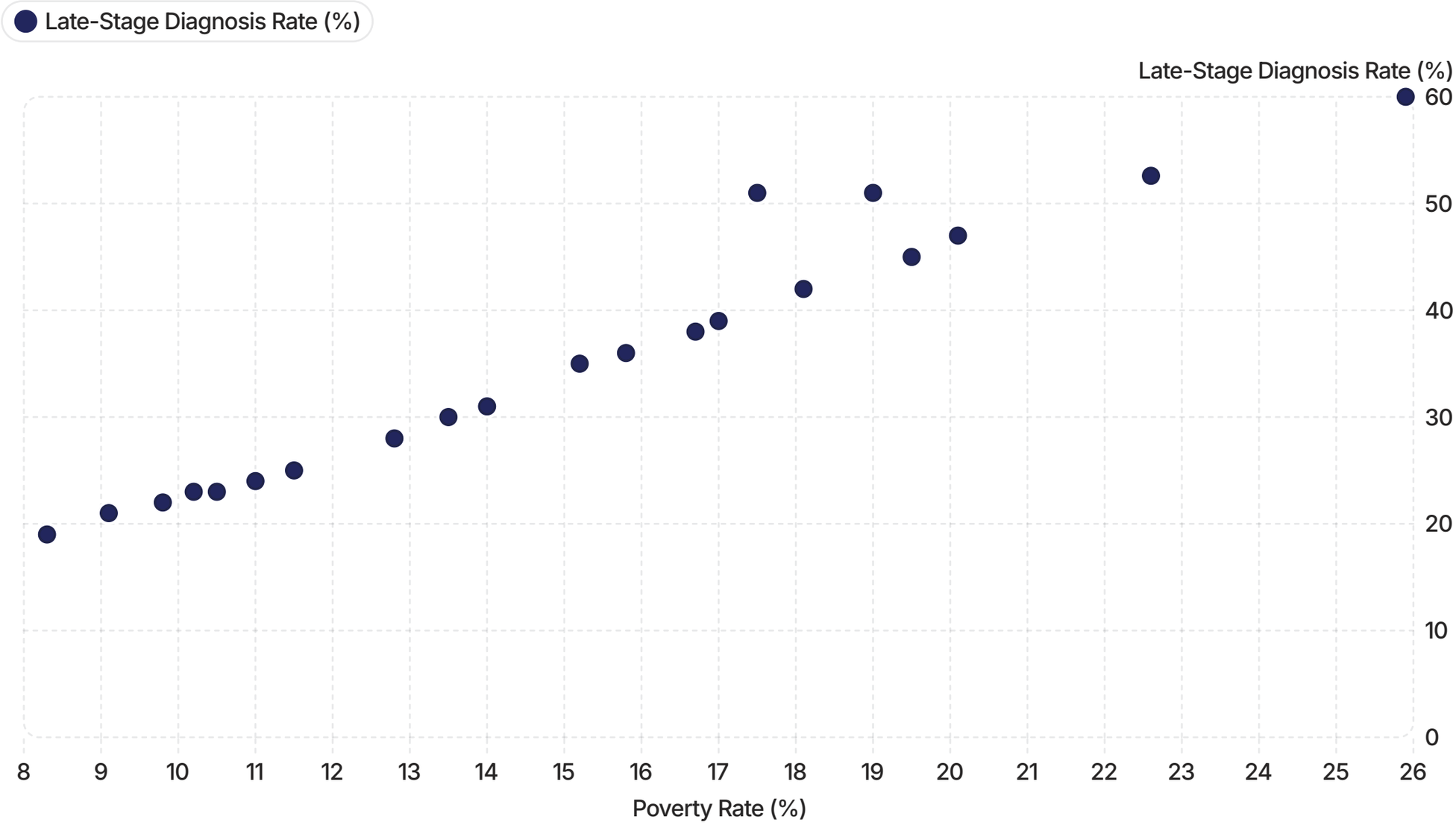
Higher Poverty, Later Diagnoses

Scatter plots illustrate the direct relationships between county-level economic factors and late-stage breast cancer diagnosis rates.



Poverty Rate vs. Late-Stage Diagnosis Rate

Counties with the highest poverty rates consistently show the most late-stage diagnoses.

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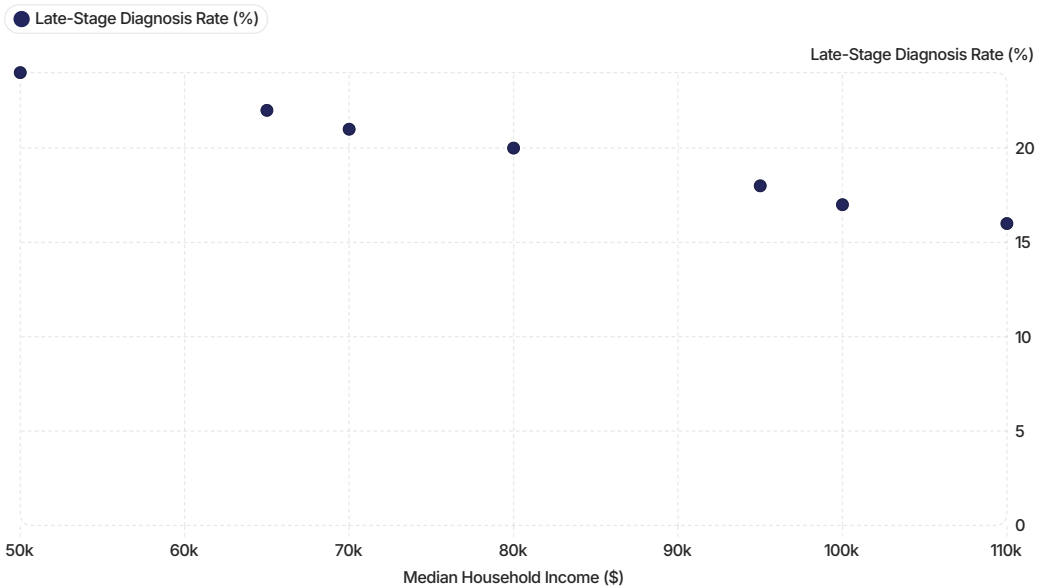
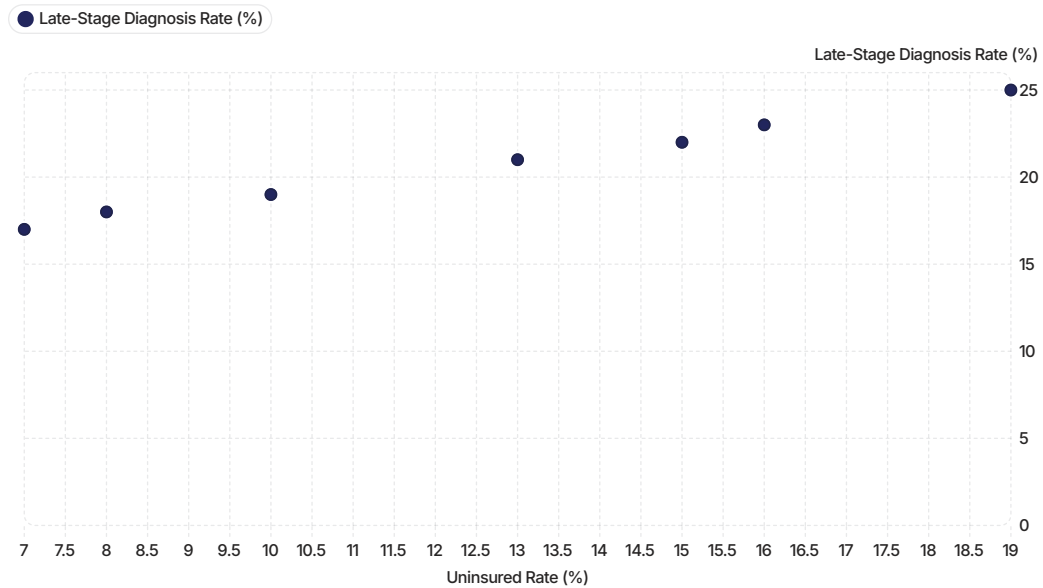
Income & Insurance paint the Same Picture

Scatter plots illustrate the direct relationships between county-level economic factors and late-stage breast cancer diagnosis rates.



Median Household Income vs. Late-Stage Diagnosis Rate

Higher income = earlier detection. The downward trend shows that counties with higher median household incomes generally exhibit lower late-stage diagnosis rates.



Uninsured Rate vs. Late-Stage Diagnosis Rate

More uninsured = more late-stage diagnoses. The upward trend indicates that higher uninsured rates correlate with increased late-stage diagnosis rates.

Whether measured by income or insurance, economic disadvantage is linked to delayed diagnosis.

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High-Risk Counties & Rural Disparities

Four counties in the top 10th percentile for late-stage diagnosis rates, all with above-average poverty levels:

1

Merced County

60% late-stage diagnosis

25.9% poverty rate

2

Madera County

52.6% late-stage diagnosis

22.6% poverty rate

3

Shasta County

51% late-stage diagnosis

19% poverty rate

4

Yolo County

51% late-stage diagnosis

17.5% poverty rate

Rural counties show higher median late-stage diagnosis rates compared to urban counties, though many rural counties had suppressed data due to small populations, limiting this comparison.

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Final Recommendations

Simple actions that can help more women get diagnosed earlier.

Target the Hardest-Hit Counties

Focus resources on Merced, Madera, Shasta, and Yolo first.

Bring Screenings Closer to Home

Use mobile mammography vans to reach women in rural areas.

Expand Health Insurance Access

Connect more low-income women to free or low-cost coverage like Medi-Cal.

Track Progress Every Year

Measure what's working by county so resources go where they're needed most.

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